Cheat sheet

Pandas & NumPy

Data wrangling commands you will use every day

NumPy

```python
import numpy as np
a = np.arange(12).reshape(3, 4); a.shape, a.T, a[:, 1], a[a > 5]
np.mean(a, axis=0); np.where(a > 5, 1, 0); np.concatenate([a, a]); a @ a.T
```

Broadcasting	Arrays of different shapes combine when trailing dimensions match or are 1

Pandas: load and inspect

```python
df = pd.read_csv("f.csv", parse_dates=["date"])
df.head(); df.info(); df.describe(include="all"); df.shape; df.dtypes; df.nunique()
```

Select and filter

Columns	`df["a"]`, `df[["a", "b"]]`
Rows by label / position	`df.loc[rows, cols]`, `df.iloc[0:5, 1:3]`
Boolean filter	`df[(df.a > 1) & (df.b == "x")]`, `df.query("a > 1 and b == 'x'")`
Membership	`df[df.city.isin(["Chennai", "Pune"])]`

Clean

Missing	`isna().sum()`, `fillna(v)`, `dropna(subset=[...])`, `interpolate()`
Duplicates	`duplicated()`, `drop_duplicates(subset=[...], keep="last")`
Types	`astype("int")`, `pd.to_datetime()`, `pd.to_numeric(errors="coerce")`

Strings	`.str.lower()`, `.str.strip()`, `.str.contains("x")`, `.str.extract(r"(\d+)")`
Rename / drop	`rename(columns={...})`, `drop(columns=[...])`

Transform and aggregate

python

```python
df.groupby("city").agg(orders=("id", "count"), revenue=("amt", "sum"))
df.pivot_table(index="city", columns="month", values="amt", aggfunc="sum", fill_value=0)
df.merge(other, on="id", how="left")      # inner / left / right / outer
pd.concat([df1, df2], ignore_index=True)
df["rank"] = df.groupby("city")["amt"].rank(ascending=False)
df["roll7"] = df["amt"].rolling(7).mean()
df.melt(id_vars="id"); df.explode("tags"); df.sample(frac=0.1, random_state=1)
```

> ⚠ **Watch out**
>
> Avoid row loops; use vectorised operations. Use `.copy()` after filtering to avoid `SettingWithCopyWarning`.